

```
import pandas as pd
import numpy as np
from sklearn import linear_model
import matplotlib.pyplot as plt
```

```
# Load Dataset
data = pd.read_excel("/content/EMP (1).xlsx")
from sklearn.preprocessing import LabelEncoder
#print(data)
```

```
linreg = linear_model.LinearRegression()
linreg.fit(data[['Experience']], data[['Salary']])
print(linreg.coef_)
print(linreg.intercept_)
p = linreg.predict([[8]])
print(p)
data.loc[3, 'Experience'] = 8
data.loc[3, 'Salary'] = p[0][0]
```

```
[[2546.07508532]]
[23650.51194539]
[[44019.11262799]]
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does
warnings.warn(
/tmp/ipykernel_2318/2639907418.py:8: FutureWarning: Setting an item of incompatible dtype is
data.loc[3, 'Salary'] = p[0][0]
```

```
#fig, axes = plt.subplots(1, 3, figsize=(10, 5))
plt.scatter(data['Experience'], data['Salary'])
#plt.set_title('Experience vs Salary')
#plt.set_xlabel('Experience')
#plt.set_ylabel('Salary')
plt.plot(
    data['Experience'],
    linreg.predict(data[['Experience']]),
    color='blue'
)
#plt.plot(data['Experience'], linreg.predict(data[['Experience']]), color='blue')
#plt.tight_layout()
plt.show()
```

